First, a simple-looking but incorrect proof:

(Incorrect) Proof: Let g be differentiable at a and f differentiable at $g(a)$. Then

$$\begin{aligned}
(f\circ g)'(a) &= \lim_{x\to a}\frac{f(g(x))-f(g(a))}{x-a}\\
&= \lim_{x\to a}\left[\frac{f(g(x))-f(g(a))}{(x-a)(g(x)-g(a))}\cdot(g(x)-g(a))\right]\\
&= \lim_{x\to a}\left[\frac{f(g(x))-f(g(a))}{g(x)-g(a)}\cdot\frac{g(x)-g(a)}{x-a}\right]\\
&= \lim_{x\to a}\frac{f(g(x))-f(g(a))}{g(x)-g(a)}\cdot\lim_{x\to a}\frac{g(x)-g(a)}{x-a}\\
&= \lim_{g(x)\to g(a)}\frac{f(g(x))-f(g(a))}{g(x)-g(a)}\cdot\lim_{x\to a}\frac{g(x)-g(a)}{x-a} \qquad (\text{since } g(x)\to g(a) \text{ as } x\to a)\\
&= f'(g(a))g'(a),
\end{aligned}$$

where we have used that $g(x)\to g(a)$ as $x\to a$ since g is differentiable and hence continuous at a. □

1.) What is wrong with the above proof? (*Hint:* As an extreme example, consider the case when g is a constant function. [1])

[1] As another example, the function

$$g(x)=\begin{cases}x^2\sin(\frac{1}{x}) & \text{if } x\neq 0,\\ 0 & \text{if } x=0\end{cases}$$

is differentiable at 0, but has a zero on every open interval containing 0.

Now we will give a correct proof of the Chain Rule.
First, since we have two functions f and g, it will help to change notation a little bit to keep things straight. Let $y = f(x)$. Instead of h, we write Δx instead, so that $\Delta y = f(a+\Delta x) - f(a)$. Note that Δx is an independent variable, while Δy is a function of Δx. In this notation, the derivative of f at a is given by

$$f'(a) = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}.$$

Define ε to be the difference between the difference quotient and the derivative:

$$\varepsilon = \frac{\Delta y}{\Delta x} - f'(a).$$

2.) Note that ε, which is a function of Δx, is not defined at $\Delta x = 0$. Compute $\lim_{\Delta x \to 0} \varepsilon$.

3.) Fill in the blank below to extend ε to a function which is continuous at $\Delta x = 0$. Slightly abusing notation, we will continue to call the function below ε.

$$\varepsilon = \begin{cases} \frac{\Delta y}{\Delta x} - f'(a), & \Delta x \neq 0 \\ & \Delta x = 0. \end{cases}$$

4.) Show that we can write $\Delta y = f'(a)\Delta x + \varepsilon\Delta x$ for $\Delta x \neq 0$.

In summary, we have shown that, for a function f which is differentiable at a, we can write, for $\Delta x \neq 0$,

$$\Delta y = f'(a)\Delta x + \varepsilon\Delta x$$

where ε is continuous at $\Delta x = 0$ with $\lim_{\Delta x\to 0}\varepsilon = \varepsilon(0) = 0$. We will use this property of differentiable functions to prove the Chain Rule.

Proof of the Chain Rule: Suppose $u = g(x)$ is differentiable at a and $y = f(u)$ is differentiable at $b = g(a)$. By the above, we can write

$$\Delta u = g'(a)\Delta x + \varepsilon_1\Delta x = [g'(a) + \varepsilon_1]\Delta x \tag{1}$$

where ε_1 is continuous at $\Delta x = 0$ with $\lim_{\Delta x\to 0}\varepsilon_1 = \varepsilon_1(0) = 0$ and

$$\Delta y = f'(b)\Delta u + \varepsilon_2\Delta u = [f'(b) + \varepsilon_2]\Delta u, \tag{2}$$

where ε_2 is continuous at $\Delta u = 0$ with $\lim_{\Delta u\to 0}\varepsilon_2 = \varepsilon_2(0) = 0$.

5.) Show that $\lim_{\Delta x\to 0}\varepsilon_2 = 0$. (*Hint*: Use equation (1).)

6.) Substitute the expression for Δu from Equation (1) into Equation (2) to show that we can write

$$\frac{\Delta y}{\Delta x} = [f'(b) + \varepsilon_2][g'(a) + \varepsilon_1].$$

7.) Multiply out the right hand side above and compute $\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$ to complete the proof.